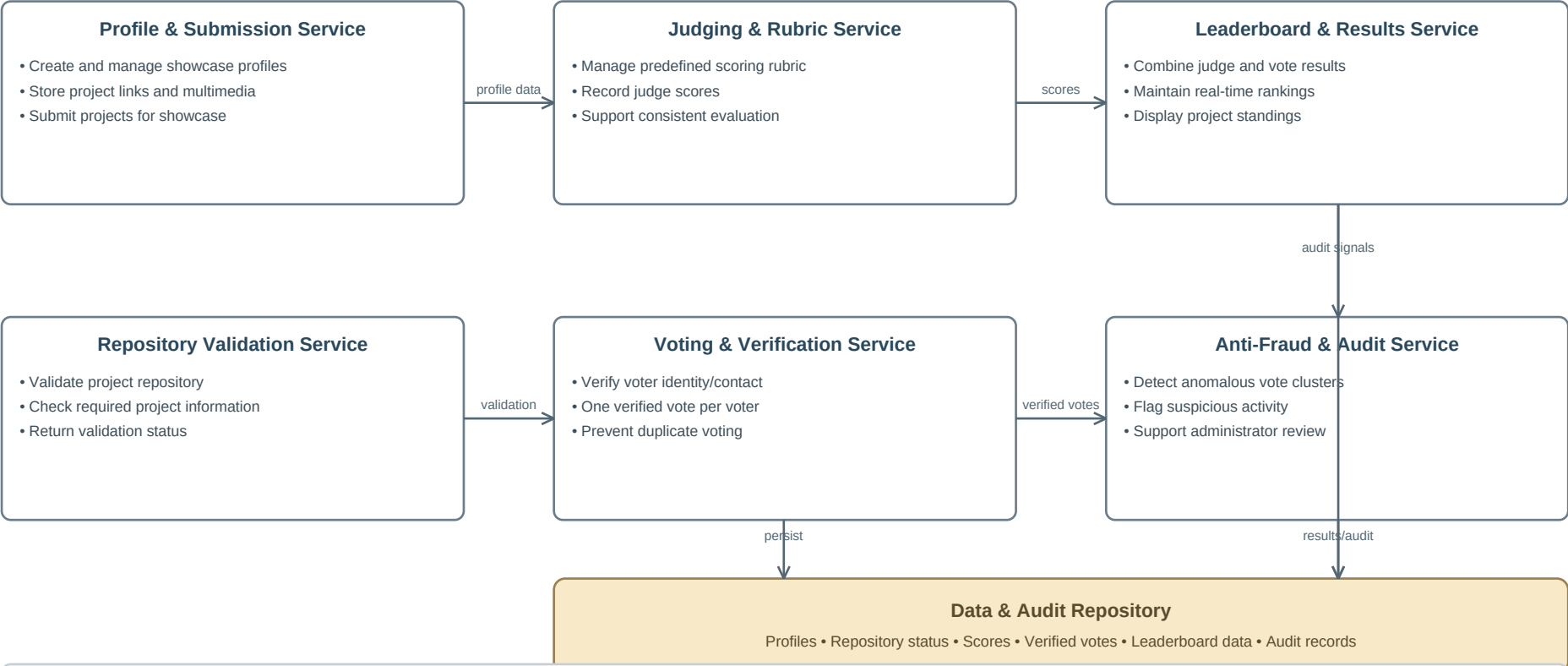


Lab 3: Component Modelling & Architectural Pattern Selection

Student Project Portfolio & Showcase App

Name: SHESANTH K | SRN: PES1UG24AM261

STUDENT PROJECT PORTFOLIO & SHOWCASE SYSTEM



Legend: Component

Interface / data flow

All services communicate through defined interfaces; no external stick-figure actors are used.

Figure 1. UML Component Diagram — Student Project Portfolio & Showcase App

Architecture Selection & Written Justification

Student Project Portfolio & Showcase App

Name: SHESANTH K | SRN: PES1UG24AM261

Selected Architectural Pattern: Layered Architecture

Presentation / API → Application Services → Validation, Judging & Voting Logic → Data & Audit Repository

Reason 1 — Clear Separation of Responsibilities

The system separates profile and submission management, repository validation, judging, public voting, leaderboard calculation, and suspicious-vote review. A layered structure keeps these responsibilities distinct while maintaining a clear processing flow.

Reason 2 — Maintainability and Clarity

Each major function can be implemented as a focused service with defined interfaces. This makes the architecture easier to understand, maintain, and modify when showcase rules, judging criteria, or voting requirements evolve.

Reason 3 — Supports Verified Public Voting

The problem statement requires one verified vote per voter and describes verification using voter identity/contact information, a one-time password, and a captured device identity or fingerprint. Separating verification from voting makes these checks explicit before acceptance.

Reason 4 — Supports Judging and Configurable Scoring

The judging requirements call for predefined scoring criteria and comparable evaluation of submissions. A dedicated judging and rubric service can manage the criteria and persist submitted scores before they are combined into results.

Reason 5 — Real-Time Leaderboard and Reliability

The requirements include real-time leaderboard updates and a target latency below 300 ms under simulated peak load. Separating leaderboard/result operations from submission and verification workflows allows each area to be optimized independently.

Security Advantage

The architecture isolates verification and suspicious-vote detection from the public showcase experience. The requirements also include anomaly detection and an administrator review path that does not block legitimate individual voters.

Performance Benefit

Separating read-heavy leaderboard/result operations from submission, validation, and voting workflows supports independent optimization and aligns with the stated real-time update and sub-300 ms response target.